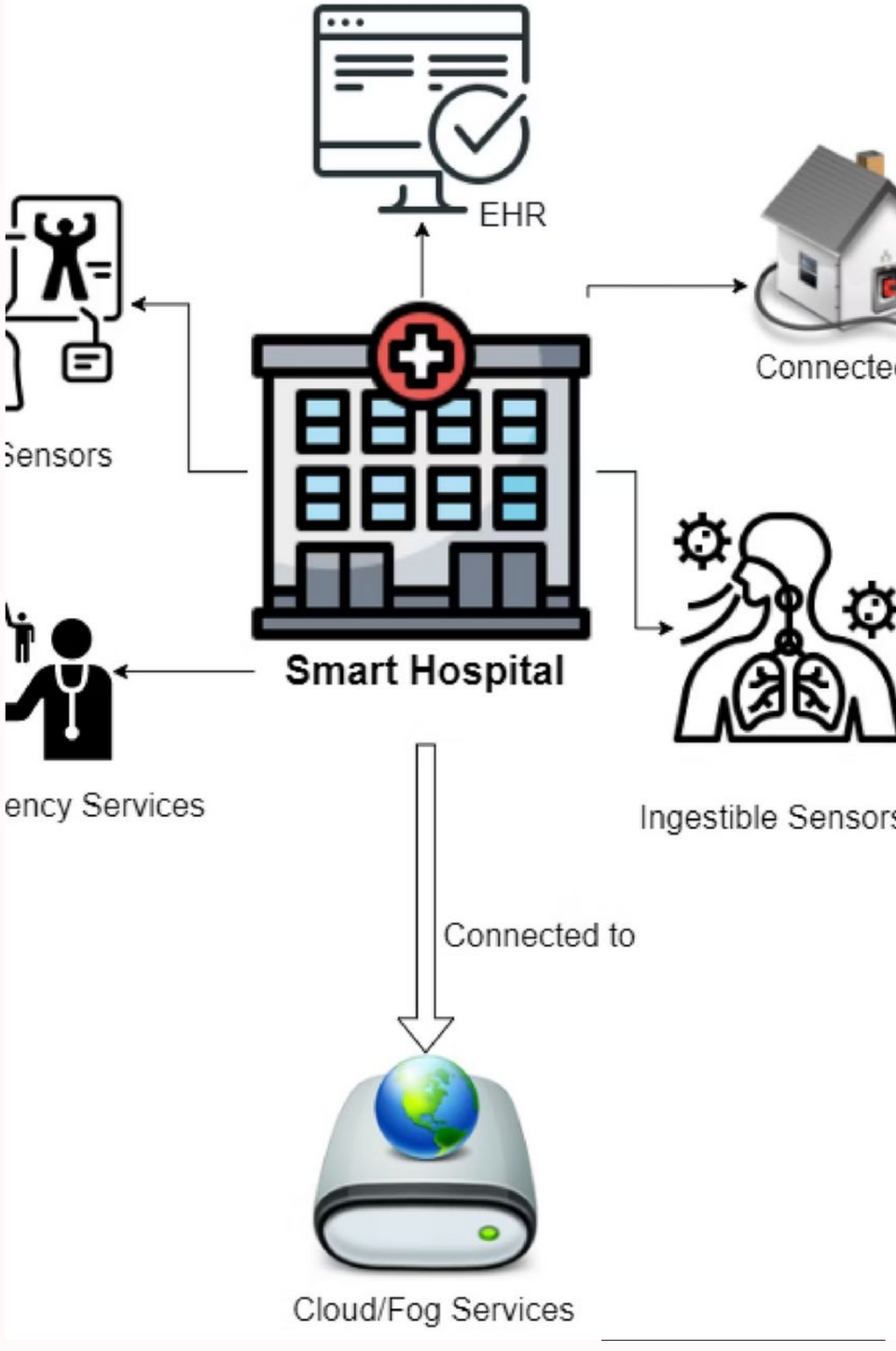
Smart Medibox

A Smart Medicine Reminder and Monitoring System

Combining intelligent reminders with environmental monitoring to ensure medication adherence and safe storage



Project Overview

The Smart Medibox is an innovative IoT device designed to help patients maintain proper medication schedules while ensuring medicines are stored in optimal conditions.

The system provides **automated reminders three times daily** — morning, afternoon, and evening — ensuring users never miss a dose.

Beyond reminders, it continuously monitors **temperature and humidity levels** inside the storage compartment, protecting medications from environmental damage that could compromise their efficacy.

Key Features

- Automated time-based reminders
- Real-time environmental monitoring
- Instant alerts for unsafe conditions
- Physical intake detection
- Web-based tracking interface

Components Used

NodeMCU ESP8266

Main controller that orchestrates all sensors, processes data, and enables Wi-Fi connectivity for web integration

DHT22 Sensor

High-precision digital sensor for measuring temperature and relative humidity inside the medicine storage compartment

DS1307 RTC Module

Real-time clock that maintains accurate timekeeping for medication reminders, even during power interruptions

Buzzer

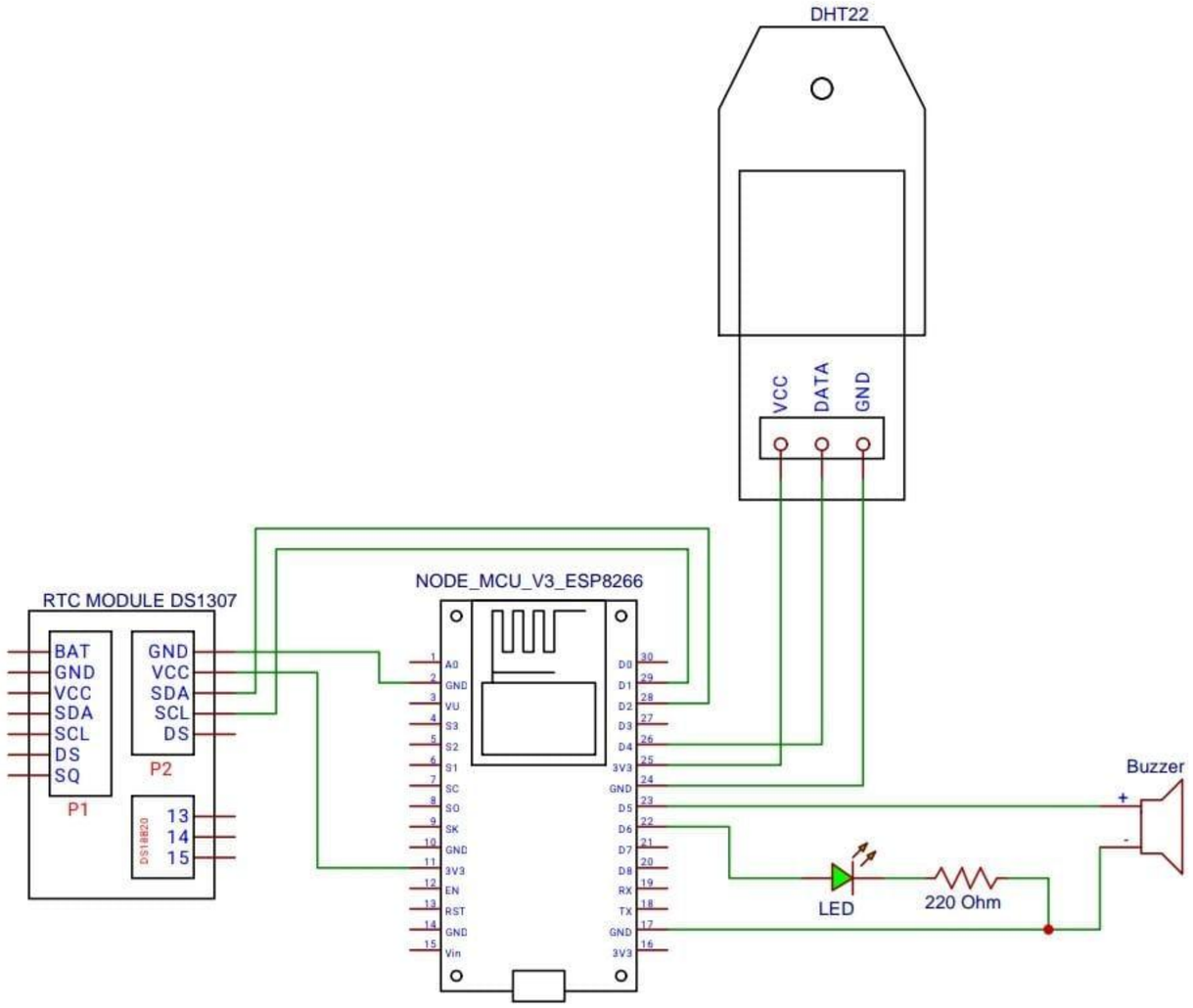
Audio alert system for medicine reminders and environmental warnings

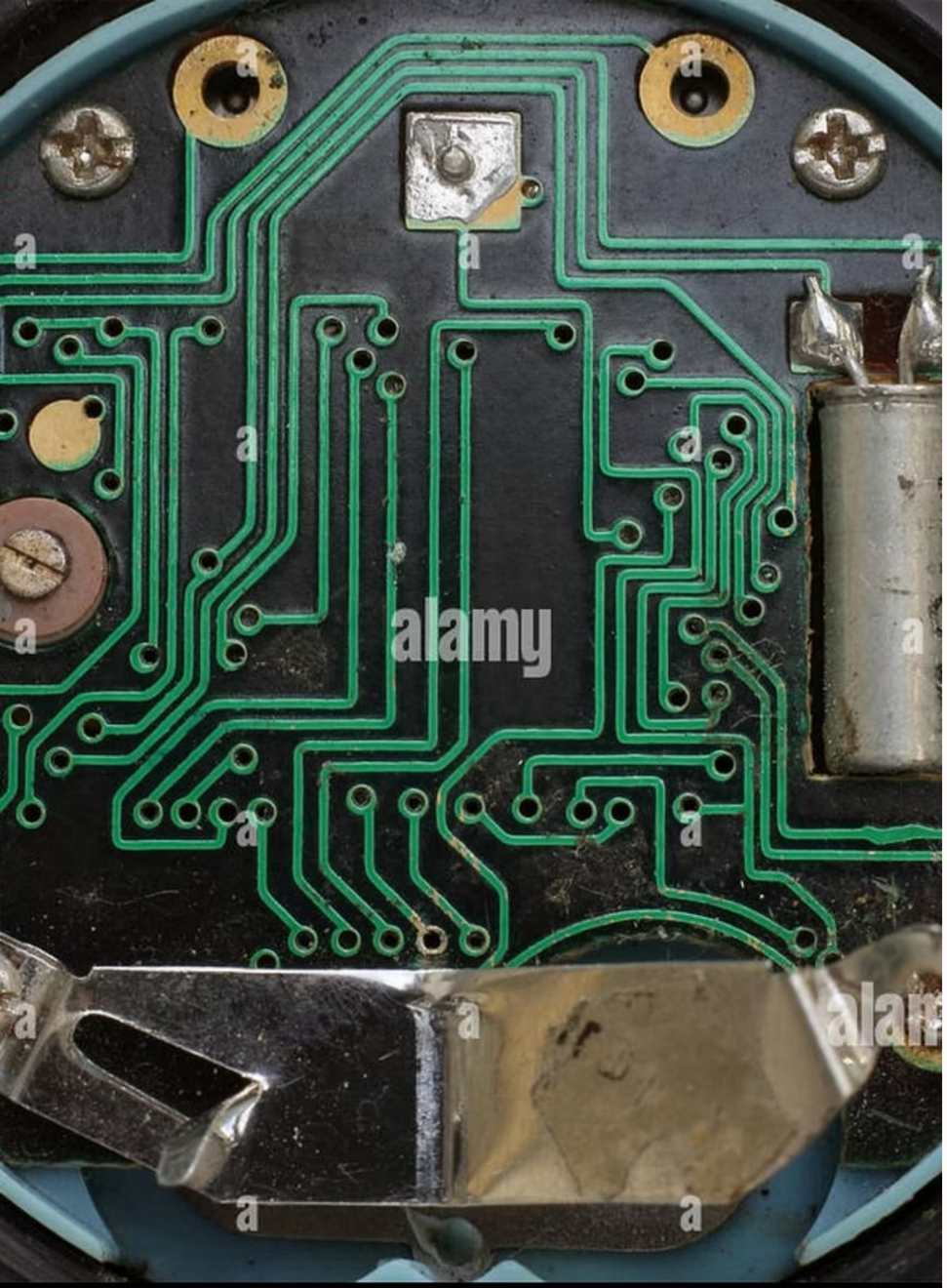
Laser & LDR Pair

Optical detection system that identifies when the medicine box is opened

LED Indicator

Visual notification system that complements audio alerts


DHT22
VCC
DATA
GND
RTC MODULE DS1307
BAT
GND
VCC
SDA
SCL
DS
SQ
P1
GND
VCC
SDA
SCL
DS
P2
DS18B20
13
14
15
NODE_MCU_V3_ESP8266
A0
GND
VU
S3
S2
S1
SC
SO
SK
GND
3V3
EN
RST
GND
Vin
D0
D1
D2
D3
D4
3V3
GND
D5
D6
D7
D8
RX
TX
GND
3V3
Buzzer
+
-
LED
220 Ohm



Reminder System Architecture

01

Time Synchronization

DS1307 RTC module maintains precise real-time tracking independent of the main controller

02

Schedule Configuration

Preset reminder times programmed into the system (e.g., 8:00 AM, 2:00 PM, 8:00 PM)

03

Alert Activation

Controller triggers buzzer and LED when current time matches scheduled reminder time

I²C Communication Protocol

- **SDA (Serial Data Line):** Bidirectional data transfer between controller and RTC
- **SCL (Serial Clock Line):** Synchronizes timing for reliable data exchange

I²C is widely used in embedded systems and MPMC (Microprocessor and Microcontroller) applications for efficient, multi-device communication with minimal wiring

Environmental Monitoring System



Continuous Protection

The DHT22 sensor performs **continuous real-time monitoring** of internal storage conditions, sampling environmental parameters at regular intervals.

When temperature or humidity values exceed predefined safe thresholds, the system immediately activates both **audio (buzzer) and visual (LED) alerts** to notify users of potentially harmful conditions.

Temperature Control

Maintains optimal storage temperature range to prevent medication degradation from heat exposure

Humidity Regulation

Monitors moisture levels to protect medicines from humidity-induced breakdown or contamination

Immediate Alerts

Dual notification system ensures users are promptly informed of any environmental risks



Medicine Intake Detection

Box Closed

Laser beam continuously illuminates the LDR photoresistor, producing a **high voltage signal**

Box Opened

Beam interruption causes **low signal**, triggering immediate buzzer activation

Intake Confirmed

Buzzer remains active until box closure, physically confirming medicine was accessed

This optical detection mechanism provides **reliable physical verification** that the medication box was actually opened during reminder periods. The laser-LDR pair creates a fail-safe system that distinguishes between acknowledged reminders and actual medicine retrieval, ensuring accurate adherence tracking.

1. Medicine Intake Detection

A **laser–LDR pair** detects whether the medicine box has been opened.

When the box is **closed**, the laser light falls directly on the LDR → **High signal**.

When the box is **opened**, the beam is interrupted → **Low signal**, triggering a buzzer to confirm medicine access.

2. Spoilage Prevention Feature

When the box remains **open for too long**, the **laser–LDR system alerts the user** through a buzzer or LED.

This prevents **exposure of medicines to air and humidity**, reducing the chance of **spoilage or contamination**.

Helps maintain a **controlled environment** for medicine storage.

Web Integration & Data Tracking

Connected Healthcare

The integrated **ESP8266 Wi-Fi module** transforms the Medibox into a fully connected IoT device, enabling seamless communication with cloud-based web servers.

After taking medication, users access a **secure web interface** to log and confirm their intake, creating a comprehensive digital record of medication adherence.



Routine Tracking

Online Data Access
The system can **send and receive data** such as reminder times, temperature, humidity, and medicine intake status.
Ensures remote accessibility and monitoring.

Medicine Intake Confirmation
After taking the medicine, the **user logs into a web interface** to mark the dose as taken.

Future scope

Add dashboard for **daily logs and environmental data** (temperature/humidity).
Implement **mobile app or IoT dashboard** for real-time health tracking.

Hardware Connections Summary

Component	Pin Connections
DHT22 Sensor	VCC → 3.3V | GND → GND | Data → D4 (GPIO2)
Buzzer	Positive (+) → D5 (GPIO14) | Negative (–) → GND
LED Indicator	Anode → D6 (GPIO12) via 220Ω resistor | Cathode → GND
RTC DS1307	VCC → 3.3V | GND → GND | SDA → D2 (GPIO4) | SCL → D1 (GPIO5)
Laser & LDR	Laser Module → 5V power | LDR signal → Analog input pin

Important: The I²C protocol (SDA/SCL) requires pull-up resistors for reliable communication. Most NodeMCU boards include these internally, but verify your specific configuration.

System Operation Flow

Time-Based Trigger

RTC module monitors current time and activates buzzer alert at scheduled medication times

Environmental Monitoring

DHT22 sensor continuously samples temperature and humidity levels inside storage compartment

Condition Assessment

System evaluates sensor readings against safe threshold parameters programmed during setup

Alert Generation

If environmental values exceed limits, dual buzzer and LED warning system activates immediately

Physical Detection

Laser-LDR optical pair detects medicine box opening through beam interruption

Web Confirmation

User logs medication intake through secure web interface, completing the adherence cycle

Smart Medibox: Complete Healthcare Solution

Key Benefits

- **Improved Adherence:** Automated reminders eliminate missed doses
- **Medication Safety:** Environmental monitoring prevents degradation
- **Verifiable Intake:** Physical detection confirms actual medicine retrieval
- **Digital Records:** Web integration enables comprehensive tracking
- **Scalable Design:** IoT architecture allows future feature expansion

Impact

The Smart Medibox represents a convergence of **embedded systems, IoT connectivity, and healthcare needs**, addressing the critical challenge of medication non-adherence.

By combining multiple sensing modalities with intuitive alerts and digital tracking, this system provides **comprehensive medication management** suitable for elderly patients, chronic condition management, and complex multi-drug regimens.

